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Tilt and rotation of the footwall of a major normal fault system: Paleomagnetism of the Black Mountains, Death Valley extended terrane, California

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ABSTRACT

Paleomagnetic data have been obtained from Miocene intrusions, including crosscutting mafic and felsic dikes, and Proterozoic crystalline rocks to evaluate deformation during Miocene and younger extension and unroofing of the Black Mountains, Death Valley, California. Synrift intrusions contain a well-defined and, at the site level, well-grouped magnetization, interpreted to be of dual polarity, the *in situ* direction of which is discordant in declination and inclination with an expected late Cenozoic reference direction. The oldest of these intrusions (11.6 ± 0.2 Ma) gives a magnetite-dominated remanent magnetization of high coercivity and high laboratory unblocking temperatures (about 550 to 580 °C) that is interpreted to predate unroofing from mid-crustal depths. *In situ* site mean directions of this magnetization are directed toward the west and west-northwest with moderate to shallow positive (down) and negative (up) inclinations. The variation in direction of magnetization, particularly inclination, with site locality around the turtleback structures along the western flank of the Black Mountains, is interpreted to result from folding of the intrusion after remanence acquisition.

Younger intrusions (≤8.7 Ma) generally give magnetizations with inclinations similar to expected Miocene values. Two populations of *in situ* site means are identified: one with southwest declination and negative inclination; the other with northward declination and positive inclination.

A preferred interpretation for footwall deformation involves, from oldest to youngest: (1) southwest-side down tilting of the entire range block of some 20° to 40° to possibly 70° and, at least locally, folding of the crystalline rocks, on a trend parallel to the Death Valley turtlebacks, between 11.6 and 8.7 Ma; (2) progressive east-to-west footwall unroofing between 8.7 and ca. 6.5 Ma; and (3) clockwise rotation (50° to 80°) of much of the Black Mountains after the core detached from stable terrane to the west. We interpret late rotation of the Black Mountains as oroflexure related to right-lateral shear along the Death Valley fault zone.

INTRODUCTION

Steeply dipping (40° to 90°) Tertiary strata in many range blocks of the Basin and Range province indicate considerable upper-crustal tilting associated with detachment faulting. Determining crustal tilt in layered rocks is usually straightforward, requiring only the measurement of geologic dips. Assessment of both absolute and relative magnitudes of vertical axis rotation of crust capped by layered strata has been successfully carried out in several paleomagnetic studies using declination discordances between observed and expected values. Assessing tilt and/or rotation in mid-crustal crystalline rocks unroofed during extension, and separated from overlying layered strata by detachment faults, however, is difficult given the lack of reliable reference to the paleohorizontal datum prior to deformation. This information, however, is critical for addressing kinematic models of the origin of detachment faults, such as whether detachments are active at low angles or passively tilted into shallow orientations with extension (Buck, 1988; Wernicke and Axen, 1988; Livaccari and others, 1993).

Paleomagnetism provides one means of evaluating the magnitude and sense of tilting and/or rotation of crystalline rocks, if pre-tilt magnetizations are identified and continent-scale latitudinal translation of the rocks can be excluded (Geissman, 1990; Marquis and Irving, 1990; Beck, 1992). Young extended terranes intruded by synrift mid-crustal magmas are unique opportunities to evaluate deformation, particularly if several intrusions span the unroofing history of the range block, and if their thermal history is well understood. In this paper, we describe paleomagnetic data from crystalline rocks of the Black Mountains, Death Valley, California, a range block intruded during two major episodes of Miocene magmatism. The excellent exposure of these intrusions makes the Black Mountains a prime area for such a study. Recent U/Pb and $^{40}Ar/^{39}Ar$ thermochronologic information bears on the intrusive and thermal history of the range during extension and unroofing and aids in the interpretation of the paleomagnetic data (Asmerom and others, 1990; Holm and others, 1989, 1992). The paleomagnetic data from the Black Mountains suggest a kinematic history more complex than that previously envisioned for extension and unroofing in the Death Valley region.

GEOLOGIC SETTING

The Death Valley area (Fig. 1) is probably the most strongly extended region in the United States Cordillera. The Black Moun-

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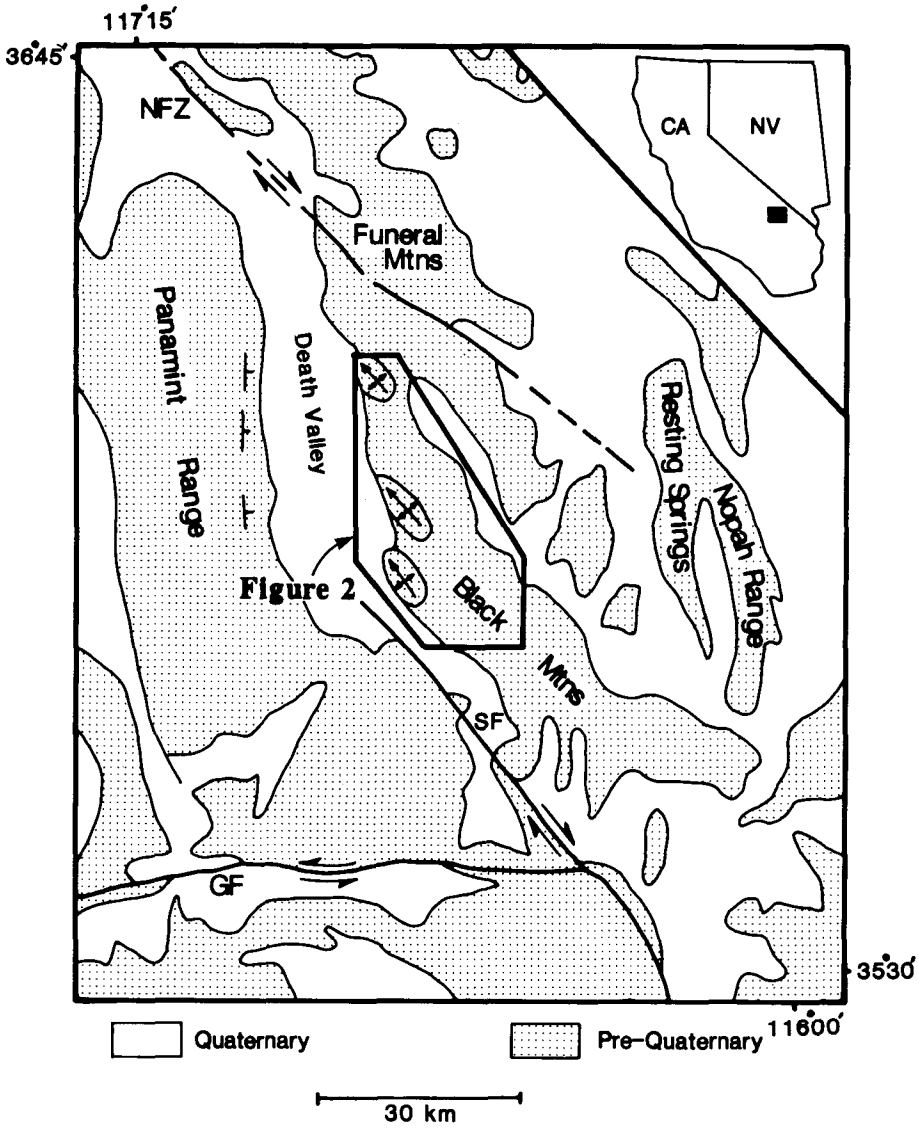


Figure 1. Index map of range blocks in the Death Valley region. GF, Garlock fault; NFZ, Northern Death Valley–Furnace Creek fault zone; SF, Southern Death Valley fault zone. Antiforms represent turtleback structures along the western flank of the Black Mountains.

tains, on the east side of Death Valley, occupy the central portion of the extended terrane, an area bounded on the east by the Spring Mountains stable block and on the west by the Sierra Nevada. Ranges in this area, including the Black Mountains, Panamint Range, and Funeral Mountains, exhibit similar structural and metamorphic features characteristic of Cordilleran metamorphic core complexes. They are mainly east-side-down tilted slivers of the miogeoclinal section whose eastern portions are unconformably overlain by moderate to steeply (>40°) east-tilted Tertiary strata and whose western portions are variably metamorphosed and ductilely deformed (Holm and Wernicke, 1989, 1990).

The crystalline core along the northwestern margin of the Black Mountains consists of 1.7 Ga basement and metamorphosed Eocambrian strata and Tertiary intrusions (<12 Ma; Asmerom and others, 1990). The Proterozoic rocks lie in the cores of three northwest-trending, *en echelon* antiforms known as the Death Valley turtlebacks (Curry, 1938; Wright and others, 1974; Figs. 1 and 2). The Proterozoic rocks are structurally overlain by large sill-like bodies that form the bulk of the Tertiary intrusions. The structurally lowest one (the Willow Spring Pluton) is a gabbro-diorite batholith exposed over a 300-km^2 area of the central Black Mountains, whose lower contact is mainly concordant to foliation in the underlying Proterozoic rocks involved in the northwest-trending folds (Otton, 1977; Wright and others, 1991). A U/Pb zircon date of 11.6 ± 0.2 Ma was obtained for the batholith by Asmerom and others (1990), which is the oldest age obtained for any intrusion in the crystalline core. Hornblende plateau $^{40}Ar/^{39}Ar$ age spectrum determinations from undeformed gabbro-diorite range from 10.0 to 10.4 Ma, indicating cooling of the pluton through about 500 °C, 1 to 1.5 m.y. after crystallization (McKenna, 1990; Holm and others, 1992). Locally the gabbro-diorite is highly deformed into a ductilely sheared L-S tectonite, in part coplanar with a mylonitic foliation in the underlying Proterozoic rocks (Holm and Lux, 1991; Pavlis, 1991). Intrusions younger than the Willow Spring Pluton have been identified in the Black Mountains by crosscutting relations and isotopic age determinations (Drewes, 1963; Otton, 1977; Holm and others, 1992). The Smith Mountain Granite (Otton, 1977; Wright and Troxel, 1984) and other intrusions are porphyritic granites and quartz monzonites intruded at about 8.7 Ma (Fig. 2). Dikes and sills of granitic, latitic, and dioritic compositions crosscut both the Willow Spring Pluton and the granitic complex. Three dikes yielded $^{40}Ar/^{39}Ar$ whole-rock and biotite age spectrum plateau dates of 7.9 Ma, 6.9 Ma, and 6.3 Ma, interpreted as crystallization dates by Holm and others (1992).

Holm and Wernicke (1990) postulated, on the basis of structural and geologic characteristics and similarities with surrounding ranges, that the Black Mountains represent a northwest-deepening section of the pre-extensional crust tilted about, in present-day coordinates, a northeast-trending axis during Miocene extension. Their interpretation is consistent with northwest-directed denudation of the Panamint Range off the Black Mountains (Stewart, 1983). The Death Valley turtlebacks and the intrusions comprising the crystalline core of the range are structurally overlain by a highly extended mass of unmetamorphosed late Precambrian strata and Miocene volcanic rocks, in part making up the Amargosa chaos (Noble, 1941; Wright and Troxel, 1984). The chaos is fragmented by numerous downward-coalescing normal faults and associated cataclasis, and is an expression of strong upper-crustal extension (Wright and Troxel, 1984). These distended and faulted rocks have been interpreted as stranded slivers of the hanging-wall block which were translated 15 to 20 km westward or northwestward (Holm and Wernicke, 1990).

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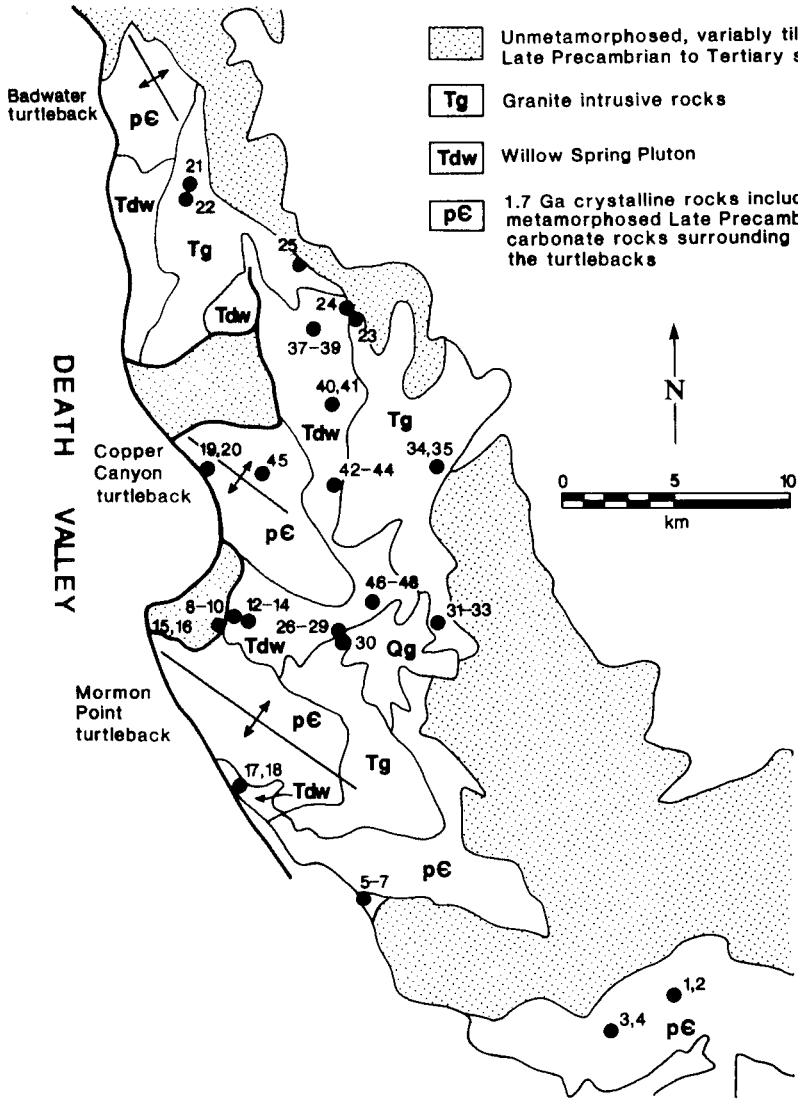


Figure 2. Index map of the Black Mountains, showing sampling locations for this study (after Drewes, 1963, and Holm and Wernicke, 1990).

Biotite $^{40}Ar/^{39}Ar$ cooling ages in the Black Mountains show that the turtlebacks were unroofed from mid-crustal depths (temperatures >300 °C) as recently as 6.8 Ma (Holm and others, 1989, 1992). The systematic distribution of biotite $^{40}Ar/^{39}Ar$ age spectrum dates throughout the range is consistent with differential cooling and, in present-day coordinates, northwest-side–up tilting about a northeast-trending axis since 11.6 Ma (Holm and others, 1989, 1992). Using the relative distribution of $^{40}Ar/^{39}Ar$ age-spectrum dates collected over an area in the central Black Mountains, McKenna and Snee (1990) proposed more than 90° west-side–up (again in present-day coordinates) tilting between 10 and 9 Ma. Paleomagnetic data allow for independent evaluation of the amount and sense of tilting and/or previously unrecognized components of vertical-axis rotation involved in the unroofing of the Black Mountains and, therefore, contribute to a better understanding of the kinematics of detachment faulting in the Death Valley region.

METHODS

A total of 476 independently oriented samples was collected with a portable gas-powered drill from 46 sampling sites throughout the Black Mountains (Fig. 2). Sample sites consisted of a single outcrop or a series of outcrops over a small area (<100 m^2). Two sampling trips were carried out to cover as much of the crystalline core as possible. Sites 1 through 35 were accessible with a 4-wheel drive vehicle or moderate hiking. Sites 37 through 48 were sampled with the aid of a helicopter.

Remanent magnetizations were measured using a 2G Enterprises Model 760R three-axis superconducting rock magnetometer in a magnetically shielded room. Alternating field (AF) demagnetizations were carried out with a 2G Enterprises AF demagnetization system. Five to twelve alternating-field demagnetization steps were performed on specimens from the majority of the samples. Thermal demagnetization was carried out using a Schonstedt TSD-1 furnace.

The results of each demagnetization run were plotted on orthogonal demagnetization diagrams. Directions of individual magnetizations isolated in progressive demagnetization and identified as linear segments defined by several demagnetization steps were calculated using principal-components analysis (Kirschvink, 1980). We rarely anchored linear segments to the origin. Magnetizations with maximum angular deviation (MAD) values greater than 15° were not included in our analysis. Some specimens possess a small magnetization (probably a viscous remanent magnetization) readily removed by treatment in either low peak fields (about 20 mT) or at low laboratory unblocking temperatures (below 300 °C). Sites, mean directions, associated statistics, and exact locations are given in Table 1. Site mean directions have been calculated following methods of Fisher (1953), giving unit weight to specimens, where usually only one specimen/sample is used in the analysis.

Anisotropy of magnetic susceptibility (AMS) data were obtained using a Sapphire Instruments SI-2 computer interfaced induction coil system. We used a 6-orientation measurement procedure, with each orientation measured twice at measurement times of about 8 sec. At the site level, mean directions of principal susceptibility eigenvectors and statistics were calculated following the method of Leinert (1991).

PALEOMAGNETIC RESULTS

Willow Spring Pluton

Samples of the Willow Spring Pluton were collected at sites in megascopically undeformed gabbro-diorite. Sample sites were selected to span as much of the exposure of the gabbro-diorite as possible, the northernmost site being located about 20 km north of the southernmost site with localities every 3 to 4 km between for a total of 18 sites. Of these,

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TABLE 1. PALEOMAGNETIC SITE DATA FROM THE BLACK MOUNTAINS, DEATH VALLEY, CALIFORNIA

Site	N	s'/s	rock	D°	I°	k	α_{95}	Long. °W, lat. °N
				Precambrian and other rocks				
1	13	11/11	pC dbd	27.4°	61.4°	63.4	5.8	116°32.92′, 35°54.56′
2	9	10/10	pC gns	13.5°	65.6°	93.6	5.0	116°32.92′, 35°54.56′
3	8	10/11	pC dbd	344.4°	53.4°	80.0	5.4	116°35.44′, 35°54.53′
4	9	11/11	fault rk	186.3°	−65.1°	86.0	5.3	116°35.44′, 35°54.53′
5	10	10/11	pC dbd	175.8°	−43.4°	28.2	9.3	116°39.88′, 35°57.25′
6	13	6/12	pC dbd	325.3°	−33.2°	46.8	8.0	116°40.17′, 35°57.69′
7	9	10/13	pC gns	11.7°	60.9°	26.2	11.8	116°40.17′, 35°57.69′
20	13	13/13	pC gns	330.7°	53.1°	50.5	5.9	116°44.13′, 36°06.61′
				Willow Spring Pluton				
8[r]	9	0/7	Tws	*	*	*	*	116°43.26′, 36°03.24′
9	9	10/13	Tws	283.1°	−20.1°	104.4	4.8	116°43.26′, 36°03.24′
10[r]	10	3/10	Tws	282.7°	−36.1°	25.2	25.1	116°43.19′, 36°03.20′
12	9	13/14	Tws	279.9°	−20.8°	110.8	4.0	116°43.19′, 36°03.20′
14	10	13/16	Tws	293.1°	−27.4°	58.0	5.5	116°43.10′, 36°03.19′
15[r]	9	*	congl.	*	*	*	*	116°43.44′, 36°03.32′
16[r]	6	*	congl.	*	*	*	*	116°43.44′, 36°03.32′
17	13	10/12	Tws	13.0°	60.5°	76.1	5.6	116°44.19′, 36°00.71′
18	13	10/10	Tws	31.3°	57.5°	69.5	5.8	116°44.19′, 36°00.71′
24	11	14/17	Tws	87.6°	−37.1°	34.1	6.9	116°40.21′, 36°10.75′
27	11	17/17	Tws	297.1°	−27.4°	106.5	3.5	116°41.27′, 36°02.96′
29	13	16/19	Tws	293.4°	−35.7°	127.6	3.3	116°41.27′, 36°02.96′
33	13	12/12	Tws	350.7°	61.7°	108.9	4.2	116°38.46′, 36°04.08′
37	12	13/13	Tws	287.4°	34.2°	26.9	9.1	116°41.02′, 36°10.15′
40	9	10/11	Tws	274.4°	9.8°	40.3	11.6	116°40.34′, 36°08.53′
41	9	13/13	Tws	92.3°	−38.4°	166.9	3.2	116°40.25′, 36°08.54′
42	9	11/13	Tws	253.7°	9.2°	90.6	4.6	116°40.10′, 36°07.07′
44	9	11/13	Tws	264.8°	−27.0°	28.0	13.5	116°40.17′, 36°07.05′
46[r]	10	2/12	Tws	281.5°	−63.3°	4000.3	3.9	116°39.71′, 36°04.54′
48	10	7/12	Tws	346.8°	54.7°	45.4	9.1	116°39.71′, 36°04.54′
				Younger intrusions				
19	10	16/16	Tpa	22.1°	63.1°	201.4	2.6	116°44.13′, 36°06.61′
13	12	11/13	Tps	1.1°	66.6°	45.8	6.8	116°43.10′, 36°03.19′
21	14	6/16	Tir	74.8°	52.7°	19.4	12.9	116°43.65′, 36°13.15′
22	11	6/16	Tir	231.2°	−38.8°	20.4	15.2	116°43.69′, 36°13.05′
23	11	6/10	Tm	161.8°	16.6°	23.3	14.2	116°40.21′, 36°10.75′
25	15	14/14	Tir	239.6°	−27.8°	27.0	7.8	116°41.04′, 36°11.67′
26	9	8/13	Tpa	253.6°	−58.2	24.4	11.4	116°41.27′, 36°02.96′
28	8	11/14	Tps	286.2°	−43.0°	24.8	9.4	116°41.27′, 36°02.96′
30	11	16/16	Tgs	326.2°	69.1°	6.0	17.6	116°40.46′, 36°02.68′
31	10	10/12	Tm	1.7°	65.6°	92.4	5.1	116°38.38′, 36°04.20′
32[r]	9	2/5	Tm	6.9°	66.5°	147.6	20.7	116°38.38′, 36°04.20′
34	12	9/15	Tm	59.0°	57.6°	37.9	8.5	116°37.78′, 36°06.56′
35	12	10/11	Tpa	252.8°	−35.9°	24.9	9.9	116°37.78′, 36°06.56′
38	6	7/8	Tpa	6.0°	58.6°	115.4	5.6	116°41.02′, 36°10.15′
39	10	8/10	Tm	357.4°	52.2°	29.4	10.4	116°41.13′, 36°10.12′
43	9	10/10	Tm	346.8°	63.1°	578.8	2.0	116°40.17′, 36°07.05′
45	10	10/11	Tgd	342.5°	55.5°	285.0	2.9	116°42.29′, 36°07.08′
47	10	8/8	Tm	0.5°	59.6°	129.0	4.9	116°39.71′, 36°04.54′

Note: N, number of independent samples collected; s'/s, ratio of number of specimens used in calculating site mean directions to total measured from N samples; D°, I°, declination and inclination of the *in situ* site mean direction; k, best estimate of Fisher precision parameter; α_{95}, half-angle of cone of 95% confidence; *, no data obtained for mean direction; [r], rejected. Rock types: Tws, Willow Spring Pluton; Tm and Tgs, Smith Mountain Granitic complex; Tpa, porphyritic andesitic dikes; Tgd, late-stage granitic dike; Tir, rhyolitic intrusive; Tps, porphyritic latite and rhyolite dikes; pC dbd, Precambrian diabase dikes; pC gns, Precambrian gneiss; fault rk, carbonate fault rock in pC dbd.

three sites were rejected after inspection of the demagnetization data and assessment of the dispersion of results. Sites 8 (sampled near a major fault; Fig. 2) and 46 gave no definable magnetization for the majority of the specimens analyzed, and site 10 gave a cone of 95% confidence radius (α_{95}) greater than our arbitrary cutoff of 18°. Cones of 95% confidence radii for the remaining 15 sites range from 3.2° to 13.5° (Table 1).

Four of the fifteen accepted sites (17, 18, 33, 48) show low coercivity (median destructive inductions less than 10 mT) and low unblocking temperatures (150–300 °C) and give a magnetization with direction subparallel to the present Earth's field. The remanence is interpreted as a viscous magnetization acquired over the past 780 ka during the Brunhes chron (Figs. 3a and 4). The remaining sites (9, 12, 14, 24, 27, 29, 37, 40–42, 44) give predominantly single-component, well-grouped magnetizations with high coercivities (median destructive inductions range from 35–70 mT; Fig. 5a), and high laboratory unblocking temperatures (550–590 °C; Fig. 5b). These temperatures exceed or are at the upper limit for the closure temperature for Ar diffusion in hornblende (McDougall and Harrison, 1988). We interpret these relations to indicate that magnetization acquisition occurred sometime between crystallization at 11.6 Ma and subsolidus cooling at 10 to 10.4 Ma. Linear trajectories of vector projections toward the origin (Figs. 6a and 6b) suggest that a stable endpoint magnetization was adequately isolated. Thermal demagnetization of representative specimens shows that, for the majority of these sites, the thermally unblocked magnetization is similar in direction to that isolated in alternating-field demagnetization. A few samples from some sites behaved in a more complicated fashion in that a stable endpoint was never reached in demagnetization, thus requiring remagnetization circles analysis for both alternating-field (Fig. 7a) and thermal (Fig. 7b) demagnetization. We interpret the well-grouped characteristic magnetization of these 11 sites as a primary thermoremanent magnetization acquired by the batholith during original cooling. For these sites, the Willow Spring Pluton shows what we interpret to be a dual polarity magnetization. The direction of this magnetization is, overall, highly discordant to an expected late Cenozoic direction (Fig. 3a). Nine sites have *in situ* site mean directions that, on average, are directed toward the west and west-northwest with moderate to shallow positive to negative (−38° to +34°) inclinations (Figs. 3a and 6a). Two sites (24 and 41) give an easterly direction with moderate negative inclination (Figs. 3a and 6b; Table 1). Note that in Figure 3a the site mean directions from sites 24 and 41 have been inverted through the origin. The reversal test on this set of data is negative according to McFadden and McElhinny (1990). The McFadden and McElhinny (1990) critical angle for the two populations is 18.8°. The angle between the mean of the normal and reverse polarity sets is 51.3°.

Site mean directions from the Willow Spring Pluton are considerably more dispersed in inclination than declination and in fact could be interpreted as two separate populations with a grand mean (of large confidence interval) of very shallow inclination. The greater dispersion in inclination is in part explained by the roughly systematic correlation between site location (with respect to turtleback structures) and inclination values. For example, sites north of the Copper Canyon turtleback give mean directions of moderate positive inclination; those to the south give moderate negative values (Fig. 2 and Table 1). Not all of the discrepancy can be associated with site location and deformation of a more local scale. For example, the mean directions from sites 42 and 44, located less than 100 m apart in the same canyon with no obvious structure in between, show an inclination difference of more than 30 degrees and are significantly different at a confidence level greater than 99.9%. We further discuss below the origin of the dispersion of site mean data as it bears on a preferred interpretation of the structural history of the intrusion.

As a test of the geologic age stability of the well-grouped, high coercivity and high laboratory unblocking temperature magnetization

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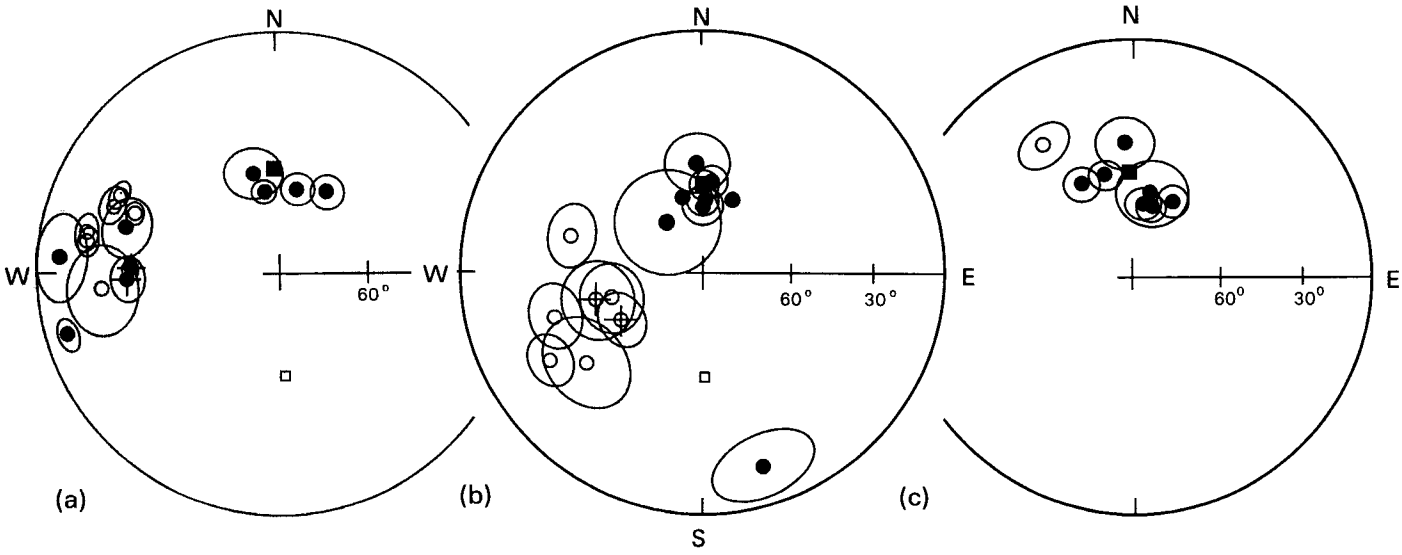


Figure 3. Equal-area projections of *in situ* site mean magnetization directions with projected cones of 95% confidence. Open symbols denote upper-hemisphere projections, solid symbols denote lower-hemisphere projections. Site mean directions inverted through the origin are indicated with a plus symbol. (a) Data from the 11.6 Ma Willow Spring Pluton; (b) data from younger granitic plutons and dikes which intrude the Willow Spring Pluton and Precambrian rocks; (c) data from Precambrian metamorphic rocks, diabase dikes, and one carbonate fault rock. Solid and open squares are Miocene expected directions (MED, normal and reverse polarity, respectively).

characteristic of most of the Willow Spring Pluton, we performed a paleomagnetic conglomerate test. We sampled cobbles of the Willow Spring Pluton at sites 15 and 16, in faulted and tilted conglomerates of Pliocene or Quaternary age, north of the Mormon Point turtleback (Otton, 1977) at the mouth of Willow Canyon (Fig. 2). In demagnetization, most cobbles yielded only single components of magnetization. The greatest percentage of the NRM was unblocked only at temperatures above 530 °C (Fig. 8a), similar to behavior exhibited by the majority of intact Willow Spring Pluton sites. For a few of the cobbles sampled, a normal polarity viscous magnetization was the first magnetization removed. For magnetizations trending to the origin and isolated over a range of high laboratory unblocking temperatures or inductions, the directions (Fig. 8b) are random at a 95% confidence level (Watson, 1956). The results of this test further support our conclusion that the well-grouped, high unblocking temperature magnetization in the gabbro-diorite was acquired during initial cooling, prior to unroofing and deformation.

Younger Intrusions

A total of 18 sites were sampled in younger intrusions. Two of these sites (30 and 32) were rejected on the basis of either an unusually low Fisher precision parameter (k = 6 for site 30) or a cone of 95% confidence

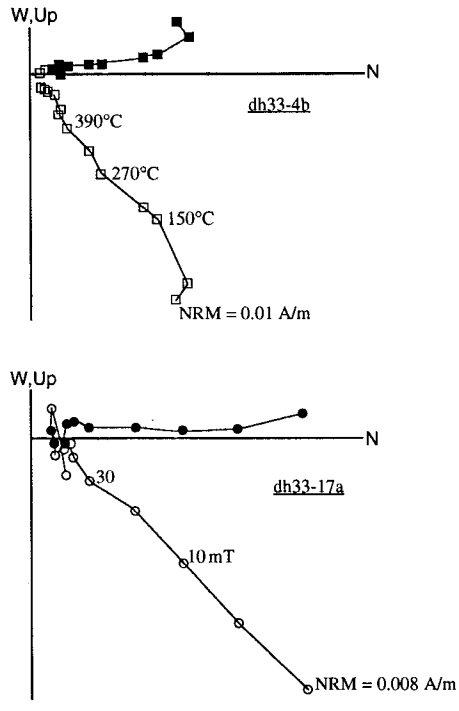


Figure 4. Orthogonal progressive demagnetization diagrams showing behavior of specimens from samples of some sites (data for site 33) in the Willow Spring Pluton during thermal and alternating field demagnetization. Solid (open) symbols represent projections of the magnetization vector onto the horizontal (vertical) plane.

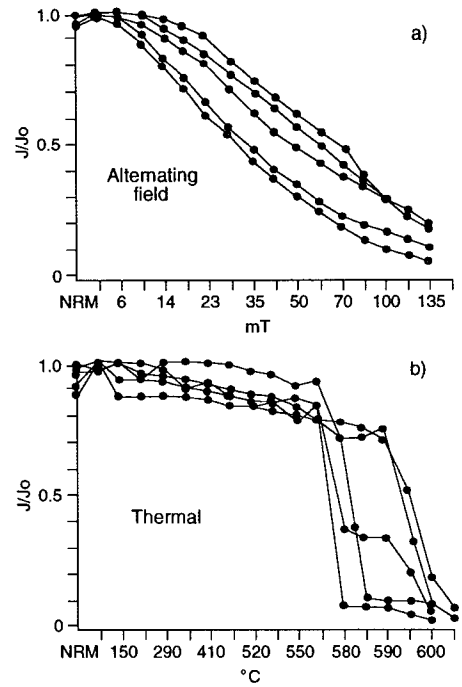


Figure 5. Normalized intensity (J/Jo) versus applied peak induction (a) and temperature (b) during demagnetization of specimens from samples representative of the majority of the sites in the Willow Spring Pluton. Note that the abscissa in each diagram is not linear.

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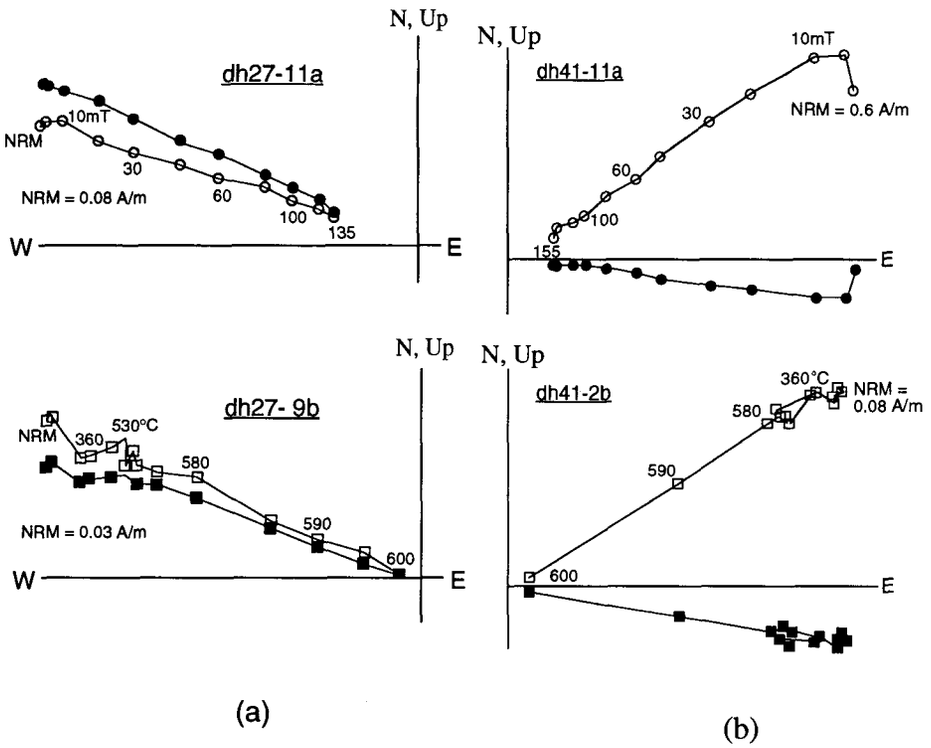


Figure 6. Orthogonal progressive demagnetization diagrams showing behavior of specimens from samples representative of the majority of the sites (data from sites 27 and 41) in the Willow Spring Pluton during alternating-field and thermal demagnetization. Solid (open) symbols represent projections of the magnetization vector onto the horizontal (vertical) plane.

greater than 18° (20.7° for site 32). Magnetizations from half of the remaining 16 sites (13, 19, 31, 38, 39, 43, 45, 47) are well grouped, yet of low coercivity and low laboratory unblocking temperature (Figs. 9a, 9b). Their directions are similar to a normal-polarity expected late Cenozoic field (Fig. 10a). The data are similar to results from four sites in the Willow Spring Pluton whose magnetizations are characterized by low coercivity and relatively low unblocking temperatures and, like them, are interpreted as post-deformation, viscous magnetizations. The remaining eight sites in the younger intrusions (21, 22, 23, 25, 26, 28, 34, 35) are well behaved with stable endpoints isolated by both alternating field and thermal demagnetization (Fig. 10b). Like the Willow Spring Pluton, these sites contain a well-grouped magnetization distinct from the expected late Cenozoic field (Fig. 3b). Five sites (22, 25, 26, 28, 35) give a magnetization of southwest and westerly declination (231° to 286°) and with moderate negative inclination (−28° to −58°). Two sites (21 and 34) give magnetizations of east-northeast declination (75° and 59°) and positive inclinations of 53° and 58°, respectively. Note that in Figure 3b the mean directions from sites 21 and 34 have been inverted through the origin. Site 23 gives an unusual direction of 162°/+17° (Table 1). The McFadden and McElhinny (1990) reversal test on the set of seven site means gives an indeterminate result. The critical angle is 21.2°; the angle between the mean of the normal and reverse polarity sets is 13.1°.

Precambrian and Other Rocks

Seven sites were selected in Precambrian rocks, the majority located in the Confidence Hills quadrangle (Wright and Troxel, 1984) in the southern Black Mountains. Three sites were in Precambrian basement gneiss (2, 7, and 20), and four were from diabase sills of probable late Precambrian age (about 1.1 Ga; Wright and Troxel, 1984). The magnetizations at most of these sites exhibit low laboratory unblocking temperatures and low coercivities (Figs. 11a, 11b). Samples from these sites, and from one site in fault rock of hematite-cemented carbonate breccia (site 4), gave an expected predominantly late Cenozoic field direction (Figs. 3c, 11c). The generally low coercivity and laboratory unblocking temperatures for magnetizations in these rocks led us to interpret them as representing a viscous magnetization, acquired after late Cenozoic uplift and deformation of these rocks.

ROCK MAGNETIC OBSERVATIONS

Rock magnetic studies were undertaken to more adequately assess the nature of the magnetization characteristic of the Black Mountains intrusions, particularly the Willow Spring Pluton, and to demonstrate that the rocks are almost certainly reliable recorders of the paleomagnetic field. Acquisition of isothermal remanent magnetization (IRM) and direct field demagnetization data from specimens of samples of representative rock types (Fig. 12) corroborate inferences, based on demagnetization behavior, that magnetite is the predominant magnetic phase in all Miocene intrusions. Coercivity of remanence values are consistently less than 60 mT. Gabbro-diorite samples from sites with magnetizations of high coercivity and laboratory unblocking temperatures show IRM acquisition curves and give coercivity of remanence values indicative of fine, single-domain and pseudo–single-domain magnetite (Dunlop, 1981). Magnetite is the dominant opaque phase present in the gabbro-diorite and petrographically appears unaltered under reflected light. For some of the Precambrian rocks sampled, IRM acquisition data and demagnetization results suggest that hematite also carries some of the remanence.

Progressive thermal demagnetization of saturation IRM (SIRM) demonstrated for all gabbro-diorite samples that SIRM unblocking temperature spectra are more distributed than those of the TRM in duplicate specimens from the same sample or specimens from adjacent samples at the same site (Fig. 13). We interpret these results to indicate that a part of the relatively coarse-grained magnetic fraction does not carry a geologically significant remanence. We see no reason to alter our argument that the characteristic magnetization of the intrusions is a thermoremanent magnetization blocked at high temperatures.

Megascopically, the Willow Spring Pluton and younger intrusions sampled at all sites appeared undeformed. For sites covering the range of intrusions sampled, we obtained anisotropy of magnetic susceptibility (AMS) data to test for the presence of a substantial magnetic anisotropy and/or well-developed magnetic fabric and to assess whether such a fabric appeared to influence magnetization acquired. Magnetic fabric data, as maximum,

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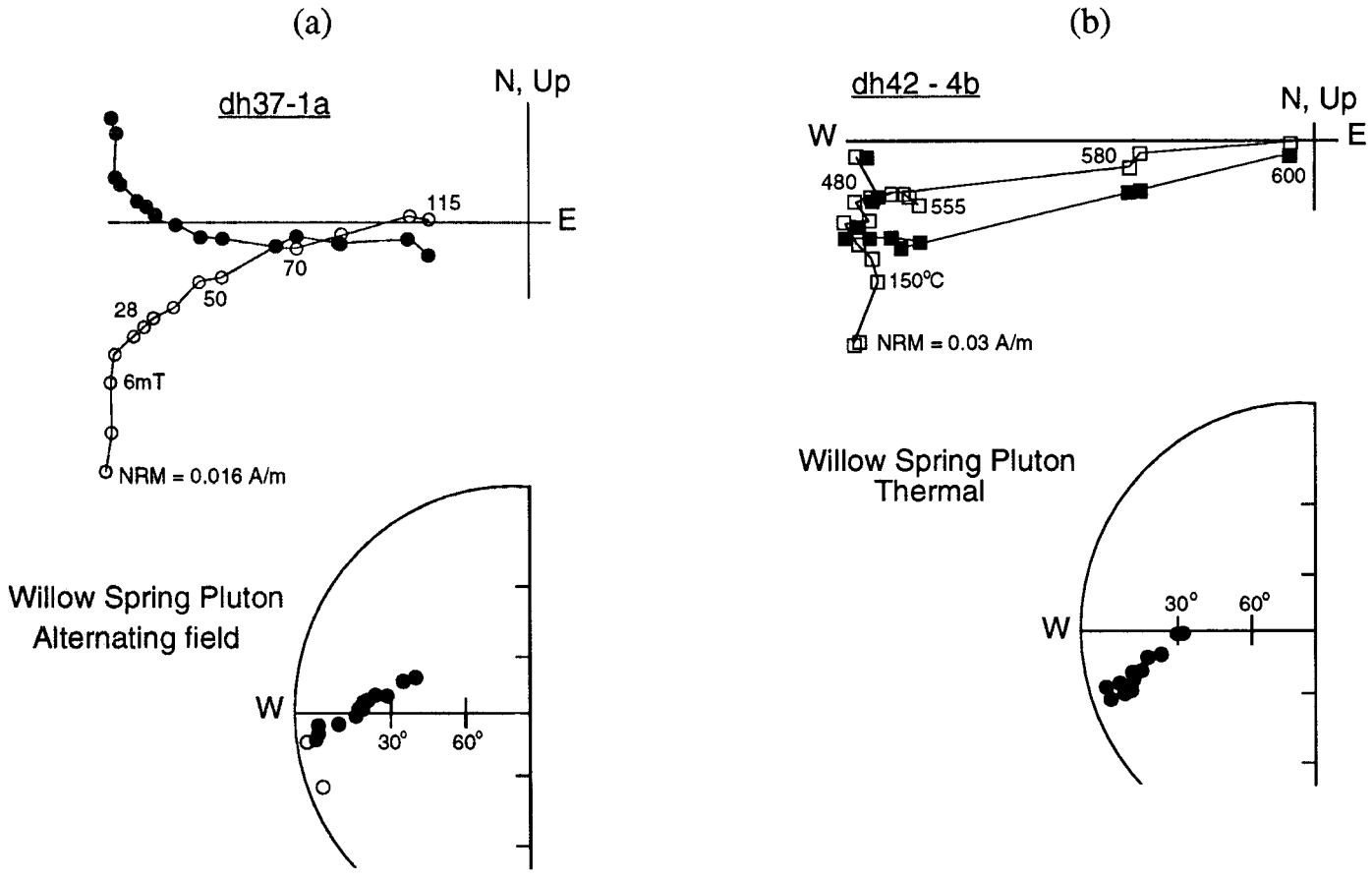


Figure 7. Orthogonal progressive demagnetization diagrams and partial equal-area projections showing examples of more complex, multicomponent demagnetization behavior, where coercivity and/or laboratory unblocking temperature spectra of magnetizations partially overlap, exhibited by samples from some sites (data from sites 37 and 42) in the Willow Spring Pluton. Solid (open) symbols denote lower- (upper-) hemisphere projections.

intermediate, and minimum principal susceptibility eigenvectors (Fig. 14), are well grouped at some of the sites, and a high degree of magnetic anisotropy is indicated by samples from sites 10, 12, 18, and 29 (Table 2). A comparison of the magnetic fabric data in Figure 14 and Table 2 and the remanence data from the respective sites (Table 1) reveals little obvious correlation among orientations of magnetic anisotropy eigenvectors, degree of magnetic anisotropy, and direction of the characteristic remanence. Such a correlation might allow us to explain the direction, as well as dispersion, of site mean characteristic magnetizations in terms of penetrative deformation of the rocks, but we emphasize that it does not appear to exist. Characteristics of the six sites for which data are shown (Fig. 14) are briefly reviewed. DH14 gives a well-defined principal magnetic susceptibility eigenvector; the direction of the site mean magnetization lies in the plane defined by the intermediate and minimum eigenvectors, but the rocks do not have a strong magnetic foliation or lineation. DH9, also with a well-defined westward and shallow negative characteristic magnetization, gives a well-defined principal eigenvector, but in this case it is vertical. DH18 gives very well-defined susceptibility eigenvectors and has a strong magnetic foliation and lineation. The site mean magnetization is well defined but is interpreted on the basis of its direction and demagnetization behavior to be of viscous origin. DH10 gives a well-defined magnetic foliation, as shown by the clustering of the minimum eigenvectors, but in this case a poorly defined characteristic direction, and the data were rejected. DH27 shows a well-defined, subhorizontal magnetic foliation and exhibits a northwest-directed lineation. The direction of the characteristic magnetization at the site lies some 50° off the foliation plane. DH29 gives a well-grouped magnetization that does not resemble any of the moderately defined susceptibility eigenvectors. The degree to which a magnetic fabric is developed at this site is variable.

DISCUSSION

Synrift Miocene intrusions from the crystalline core of the Black Mountains contain well-defined and well-grouped magnetizations, which we interpret to be of dual polarity, that are highly discordant to reverse and normal polarity expected late Cenozoic field values (178°/−55° and 358°/55°; Mankinen and others, 1987). The magnetizations, summarized in Table 3, are generally of high coercivity and high unblocking temperature and we argue that they were acquired prior to uplift from midcrustal depths (10 to 15 km). In total, the Willow Spring Pluton contains a shallow inclination magnetization (both positive and negative in sign) directed toward the west and west-northwest (and what we interpret to be opposite polarity magnetization, although the reversal test of McFadden and

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TABLE 2. ANISOTROPY OF MAGNETIC SUSCEPTIBILITY DATA, WILLOW SPRING PLUTON

Site	N	Maximum	Intermediate	Minimum	Xmax	Xmin	Xinit	F	L	E	V
8	10	258.1/55.0	98.2/33.4	1.9/9.4	1.0000	0.9904	0.9690	2.7	1.0	1.012	33.9
		57 18	57 18	29 19	0.0036	0.0044	0.0035	0.8	1.0		
9	5	254.7/78.3	35.6/9.1	306.8/−7.2	1.0000	0.9496	0.9447	3.1	5.0	0.955	72.6
		33 14	82 16	82 12	0.0263	0.0149	0.0152	3.6	4.0		
10	10	272.8/62.6	252.5/−25.9	346.5/−8.3	1.0000	0.9973	0.9345	6.6	0.0	1.064	11.8
		79 7	79 6	11 6	0.0052	0.0068	0.0064	1.2	0.0		
12	3	343.3/−10.7	78.0/−23.6	50.7/63.9	1.0000	0.9528	0.6336	38.2	5.0	1.433	21.0
		86 13	86 28	79 27	0.1342	0.0221	0.2750	35.3	16.0		
14	6	36.0/1.5	289.2/84.6	306.1/−5.1	1.0000	0.9762	0.9566	3.2	2.0	0.996	47.8
		14 6	53 12	53 6	0.0028	0.0052	0.0074	1.1	1.0		
17	9	120.6/−18.6	45.1/36.8	189.1/47.3	1.0000	0.9916	0.9641	3.2	1.0	1.020	29.0
		57 19	56 20	27 18	0.0026	0.0018	0.0017	0.4	0.0		
18	13	104.5/43.9	206.3/12.0	308.1/43.6	1.0000	0.8940	0.8195	14.1	12.0	0.975	50
		7 5	9 7	9 5	0.0175	0.0065	0.0144	2.6	2.0		
24	3	70.1/45.8	150.4/−9.5	135.0/73.1	1.0000	0.9707	0.9340	5.3	3.0	1.009	41.7
		71 4	79 31	80 63	0.0034	0.0031	0.0012	0.4	1.0		
27	6	348.8/14.1	76.5/−9.0	135.0/73.1	1.0000	0.9868	0.9694	3.5	1.0	1.015	34.8
		61 10	61 25	34 16	0.0010	0.0033	0.0041	0.6	0.0		
29	5	354.9/24.1	71.0/−28.3	118.8/51.3	1.0000	0.63431	0.6116	27.9	47.0	0.676	73.5
		7 5	45 4	45 7	0.03082	0.1554	0.1525	38.4	46.0		

Note: the second line of each site entry contains the 95% confidence errors for the magnetic susceptibility eigenvectors, and one sigma errors for the other determinations. N is number of independent, nondemagnetized samples measured. Maximum, Minimum, and Intermediate are site mean, *in situ* magnetic susceptibility eigenvectors listed as D/I, in degrees. Xmax, Xint, and Xmin are the magnetic susceptibility eigenvalues, normalized to Xmax. F is magnetic foliation, L, lineation, in percent. E is elongation factor. V is Graham's (1949) factor, in degrees.

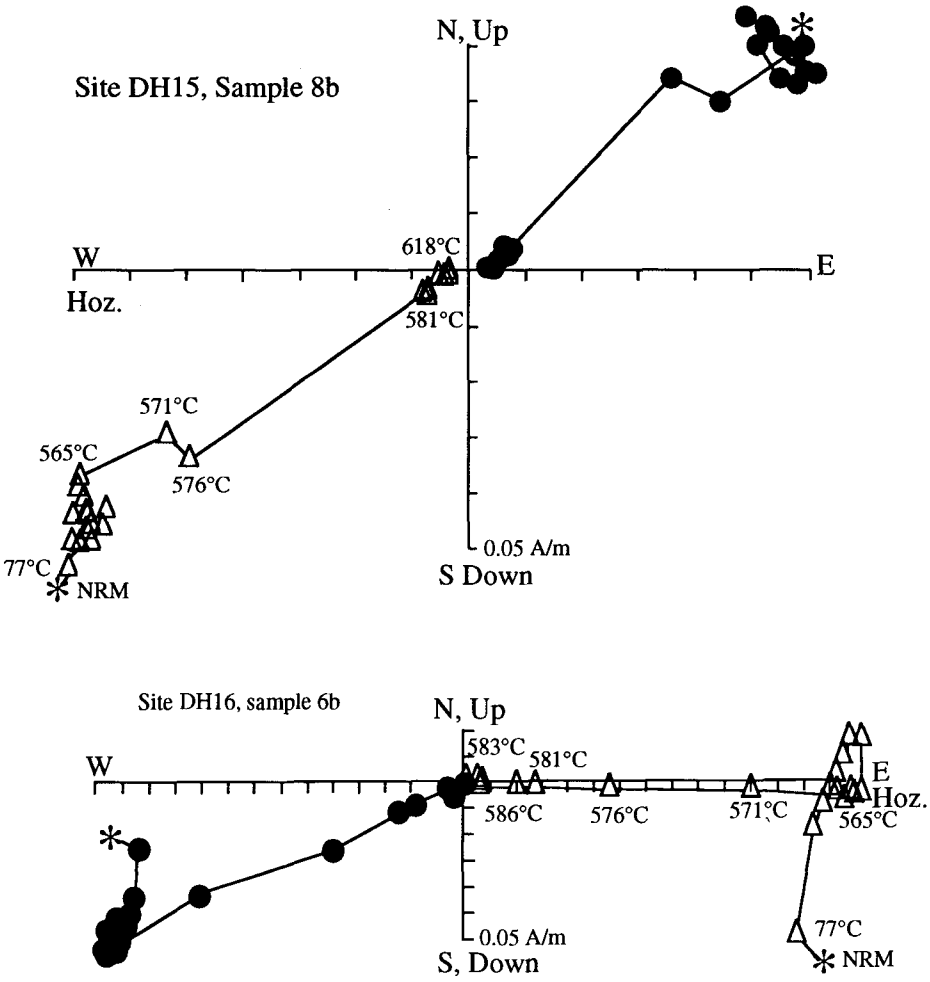


Figure 8a. Orthogonal progressive demagnetization diagrams showing typical behavior observed for specimens from samples of clasts of Willow Spring Pluton in lower Pliocene conglomerates (sites 15 and 16) during thermal treatment. Solid (open) symbols represent projections of the magnetization vector onto the horizontal (vertical) plane.

McElhinny, 1990, is negative). Magnetizations in younger intrusions are generally of steeper inclination (58° to 28°) and also show anomalous declinations directed toward the southwest and east-northeast. The paleomagnetic data for the Black Mountains are complex and can be interpreted in several ways, in part because of lack of reference to the paleohorizontal datum at the time of remanence acquisition. In the following, we argue for a sequence of events, from youngest to oldest, which we believe is most consistent with structural observations and thermochronologic data from the Black Mountains.

Evidence for Vertical-Axis Rotation

The younger intrusions in the Black Mountains (8.7 Ma and younger) give inclinations

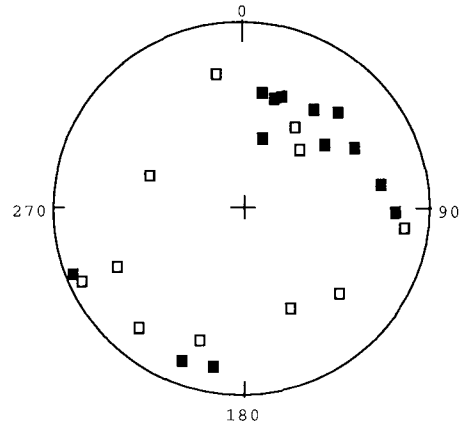


Figure 8b. Equal-area projection of directions of magnetization isolated from individual samples (one specimen/sample/clast) from conglomerate sites 15 and 16.

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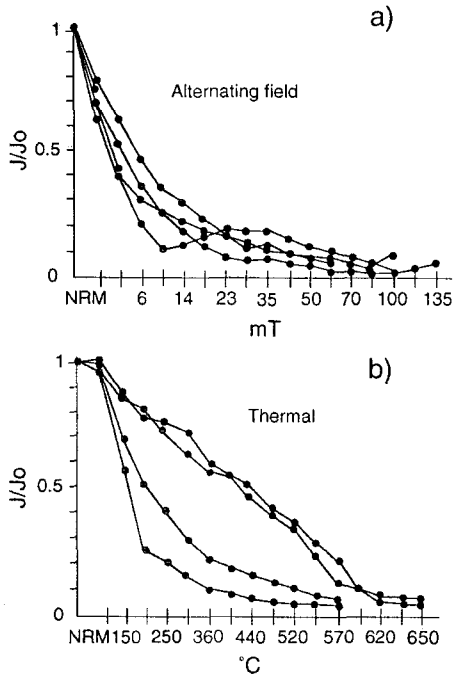


Figure 9. Normalized intensity (J/Jo) versus applied peak induction (a) or temperature (b) for specimens from samples of Willow Spring Pluton containing a single, uplift-related component of magnetization and for younger Miocene intrusions.

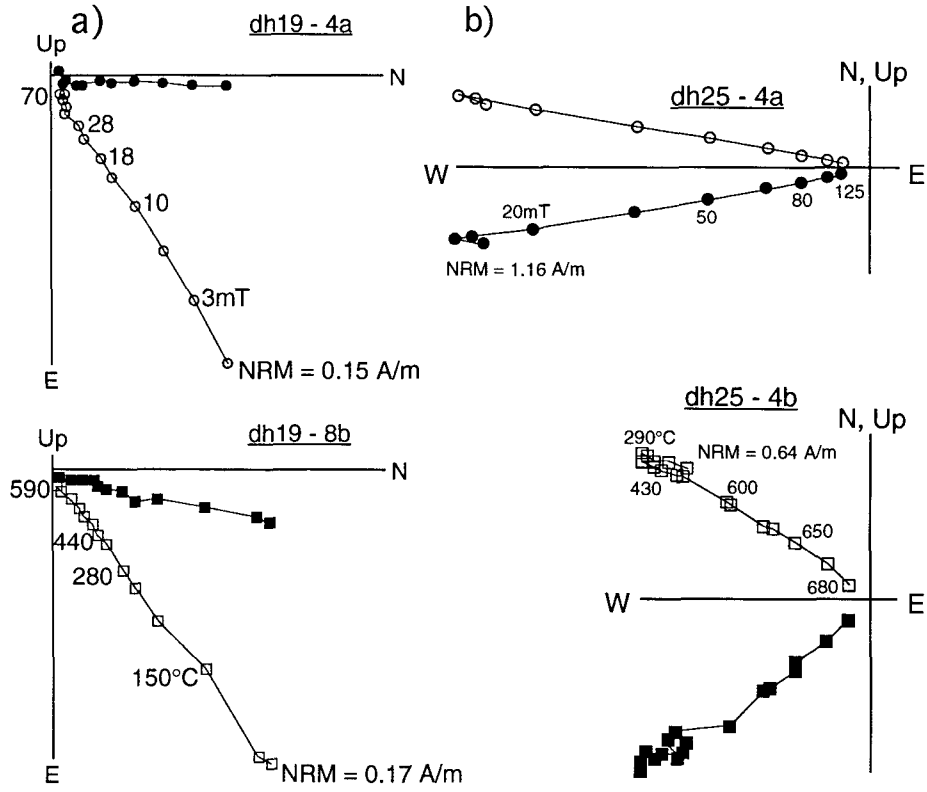


Figure 10. Orthogonal progressive demagnetization diagrams showing typical behavior observed for specimens from samples of younger Miocene intrusions during progressive alternating-field and thermal demagnetization. (a) Data from site 19; (b) data from site 25. Solid (open) symbols represent projections of the magnetization vector onto the horizontal (vertical) plane.

of characteristic magnetizations (average of 46°; Table 3) similar to an expected late Cenozoic field. The discordant declination of the overall mean can be interpreted in two (end-member) ways: (1) northwest-side–down tilting around a northeast-trending axis or (2) vertical-axis clockwise rotation. Variations in distribution of rock types across the range seem to make the first interpretation untenable. In the southeast, Tertiary volcanic rocks of pre–10 Ma age are deposited on unmetamorphosed late Precambrian–Cambrian strata; to the northwest, there are exposures of coeval mid-crustal plutons and amphibolite-facies equivalents of the late Precambrian–Cambrian strata. Field observations (Miller, 1991; Holm and Wernicke, 1990; Wright and others, 1974; Otton, 1977), $^{40}Ar/^{39}Ar$ age-spectrum determinations (Holm and others, 1992; McKenna and Snee, 1990), and regional reconstructions of Tertiary extension (Snow and Wernicke, 1989; Stewart, 1983; Wernicke and others, 1988) all indicate top-to-the-northwest denudation and thus, in present-day coordinates, *northwest-side–up* unroofing of the Black Mountains during Tertiary extension.

We suggest that the discordant *in situ* paleomagnetic data from the young intrusions record a clockwise vertical-axis rotation of some 50° to 80° of the Black Mountains crystalline core since ca. 8.7 Ma. The possibility of significant vertical-axis rotation during crustal extension and its importance for understanding the three-dimensional kinematics of extension have been suggested in both structural and paleomagnetic studies (for example, Burchfiel, 1965; Guth, 1981; Nelson and Jones, 1987). The first study to document such vertical-axis rotations paleomagnetically was that of Gillett and Van Alstine (1982) along the Las Vegas Valley shear zone. Block rotation associated with detachment faulting, or occurring shortly thereafter, has been documented in east-central Idaho (Janecke and others, 1991), west-central Nevada (Hudson and Geissman, 1987, 1991), the Mojave extensional belt of southern California (Ross and others, 1989), and in the Colorado River extensional corridor (Wells and Hillhouse, 1989). In the Death Valley region, large-scale rotation associated with Tertiary

TABLE 3. SUMMARY OF PALEOMAGNETIC DATA FROM CRYSTALLINE ROCKS OF THE BLACK MOUNTAINS

Rock	N/No	Decl.	Incl.	α_{95}	$\alpha_{951\text{-}3}$	$\alpha_{951\text{-}2}$	k	k_1	k_2
Precambrian and other rocks									
Thermoviscous, uplift-related	7/8	359.3,	58.8	9.7	5.2	9.9	39.4	−55.3	−15.5
Willow Spring Pluton									
TRM, westward declination only	9/14	280.3,	−12.6	18.1	9.9	19.3	9.0	−14.1	−4.3
TRM, total accepted sites	11/14	278.6,	−3.3	19.1	8.6	24.6	6.6	−16.6	−2.8
Thermoviscous, uplift-related	4/5	5.3,	59.9	12.9	2.6	11.6	52.0	−359.8	−19.0
Younger intrusions									
TRM, both polarities	7/7	250.5,	−46.1	13.1	8.8	12.2	22.1	−20.1	−10.8
Thermoviscous, uplift-related	9/9	356.5,	62.2	5.7	4.5	5.2	82.5	−55.8	−41.4

Note: N/No, ratio of sites accepted for statistical analysis to total number of sites collected and/or giving the magnetization of interest; Decl., Incl., declination and inclination of the *in situ* site mean direction, in degrees. α_{95}, semi-angle of the cone of 95% confidence about the site mean; $\alpha_{951\text{-}3}$, $\alpha_{951\text{-}2}$, Bingham (Onstott, 1980) approximate 95% confidence angles; k, Fisher precision parameter; k_1, k_2, maximum likelihood estimates of Bingham concentration parameters.

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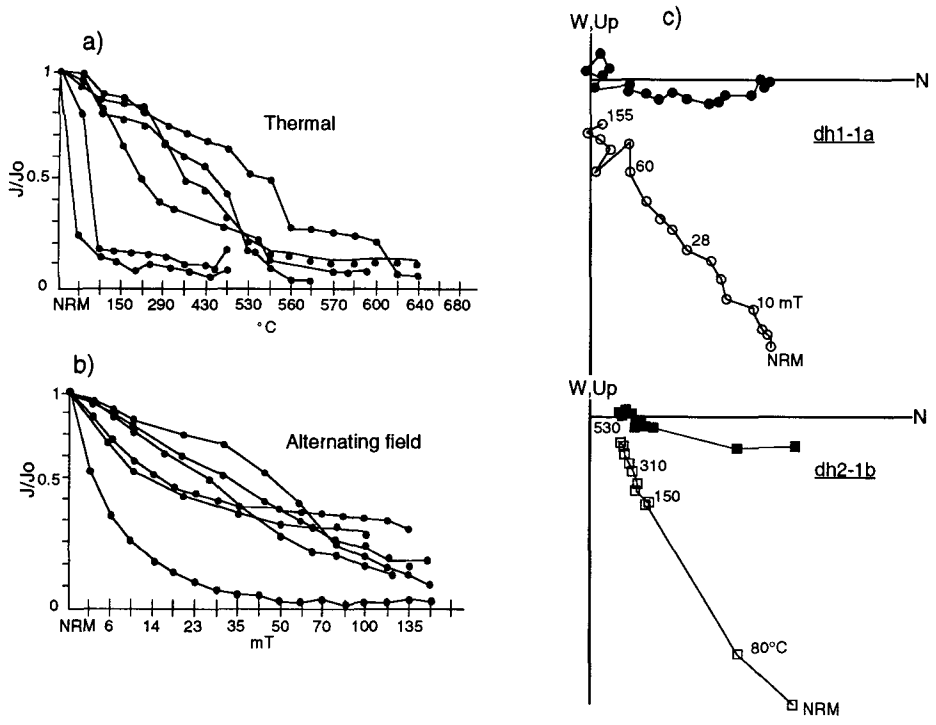


Figure 11. Normalized intensity (J/Jo) versus applied peak induction (a) or temperature (b) from specimens from samples of Precambrian rocks. (c) Orthogonal progressive demagnetization diagrams showing typical behavior observed for specimens from samples of Precambrian metamorphic rocks (site 2) and diabase dikes (site 1) during alternating field and thermal demagnetization. Solid (open) symbols represent projections of the magnetization vector onto the horizontal (vertical) plane.

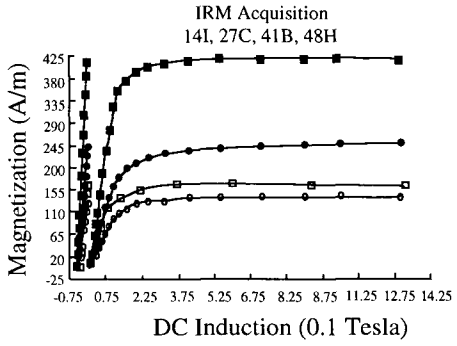


Figure 12. Curves showing the acquisition of isothermal remanent magnetization (IRM) and direct field demagnetization for specimens from four samples (14I [closed circles], 27C [open circles], 41B [closed squares], and 48H [open squares]) from sites in the Willow Spring Pluton that contain well-grouped magnetizations of high coercivity and high, narrow laboratory unblocking temperatures.

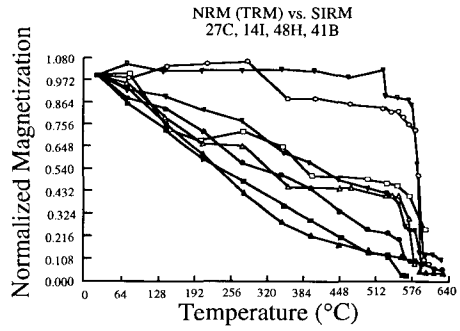


Figure 13. Curves showing the response in thermal demagnetization of the natural remanent magnetization (NRM) and saturation isothermal remanent magnetization (SIRM) for duplicate specimens from samples with IRM acquisition data shown in Figure 12. Open symbols show the NRM response; closed symbols show response of SIRM. Samples are as follows: 14I (squares), 27C (circles), 41B (upright triangles), 48H (inverted triangles).

extension has been proposed by Snow and Wernicke (1989) for other range blocks based on structural continuity of regionally correlative thrust sheets. Their reconstruction predicts about 90° counterclockwise rotation of the southern Grapevine Mountains, 60° to 90° clockwise rotation of the Bare Mountain and Striped Hills blocks, and little or no rotation for the intervening Funeral Mountains. Such large-scale rotations are consistent with recent results of paleomagnetic studies which suggest that block rotations may preferentially occur in highly extended terranes and may be heterogeneous in nature over a given region (Hagstrum and Gans, 1989; Nelson and Jones, 1987; Wells and Hillhouse, 1989; Janecke and others, 1991).

The Death Valley region has been suggested to lie within a zone of diffuse intracontinental shear (Atwater, 1970; Wernicke and others, 1988) termed the "Eastern California shear zone" (ECSZ) (Dokka and Travis, 1990). This zone, occupying much of the western Mojave Desert south of Death Valley, and possibly extending to the Walker Lane belt (western Nevada) north of Death Valley (Stewart, 1988), is characterized by tectonic domains within which relatively uniform vertical-axis rotation of the entire domain has been suggested by paleomagnetic data (Ross and others, 1989; Ross, 1991), or has been inferred on the basis of field geologic observations (Burchfiel, 1965; Albers, 1967; Dokka and Travis, 1990). Rotation of blocks within these domains, as well as their internal deformation by strike-slip faulting, has been interpreted to be the result of broadly distributed regional right shear accommodating Pacific–North American plate motion (Atwater, 1970; Wernicke and others, 1988; Calderone and others, 1990; Dokka and Travis, 1990). In the Mojave extensional belt south of Death Valley, the pattern of rotation is complex (Golombeck and Brown, 1988; Ross and others, 1989), and locally rapid rotation of the crust has been suggested (Ross and others, 1989). Given the recent recognition of vertical-axis rotation in highly extended terranes and the possible involvement of the Death Valley region in the proposed ECSZ, we interpret the observed declination discrepancy in the paleomagnetic data as representing clockwise rotation of the Black Mountains associated with extension.

Clockwise rotation of the Black Mountains is consistent with extension-related features in the Death Valley area. Although the present extension direction in the region is northwest (Burchfiel and others, 1987; Wernicke and others, 1988), a clockwise shift in extension direction in the interval 5 to 10 Ma likely occurred, as pre–10 Ma extended terranes in the southern Great Basin are predominantly east-west or southwest-northeast (Zoback and others, 1981). For the Black Mountains, slickensides, fault mullions and mylonitic stretching lineations observed in the turtlebacks along the northwest flank

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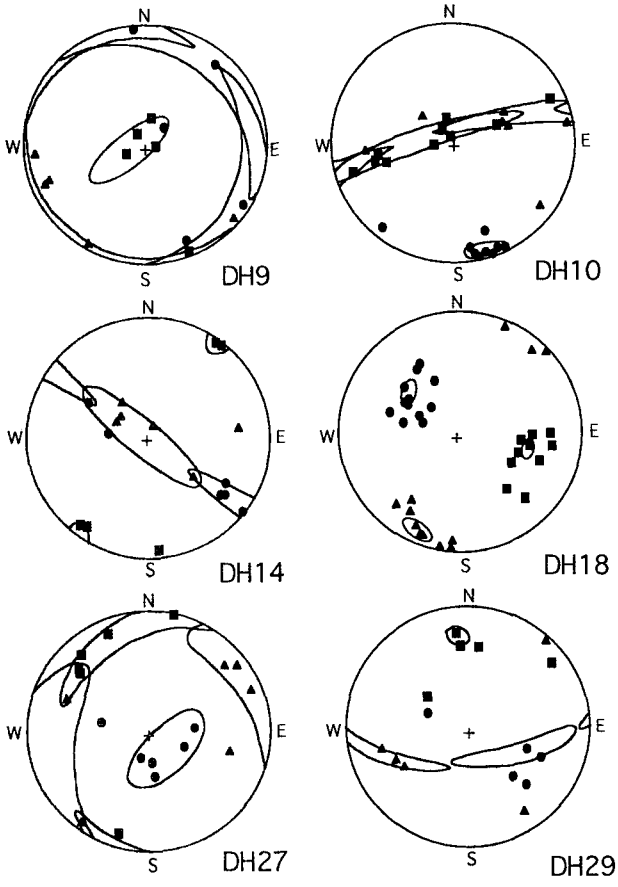


Figure 14. Equal-area, lower-hemisphere projections of specimen magnetic susceptibility eigenvectors for samples from six sites in the Willow Spring Pluton exhibiting a range of paleomagnetic behavior. In each projection, squares refer to maximum eigenvectors, triangles to intermediate eigenvectors, and circles to minimum eigenvectors. Associated statistics and magnetic fabric data are given in Table 2.

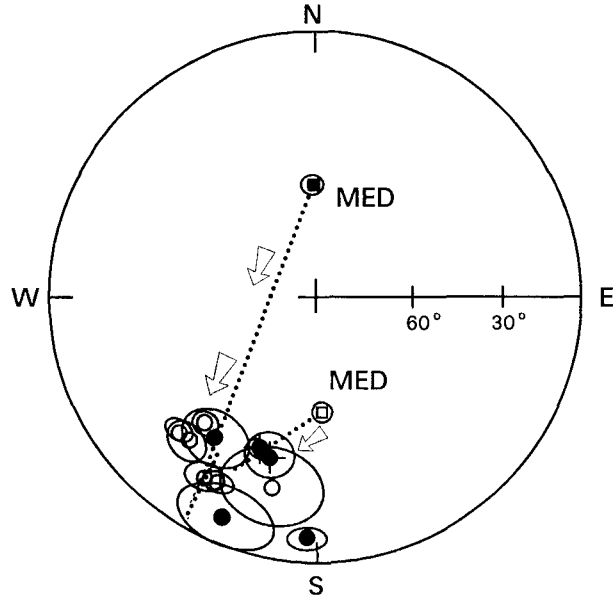


Figure 15. Equal-area projection showing partially restored site mean directions from the Willow Spring Pluton, after removal of an estimated average of 71 degrees of clockwise rotation of the studied portion of the Black Mountains. Solid and open squares are Miocene expected directions (MED, normal and reverse polarity, respectively). Dotted lines indicate two possible deformation paths, involving whole-scale tilting of the batholith about vertical axes, prior to intrusion of younger granites and dikes. Solid (open) symbols represent lower- (upper-) hemisphere projections. Site mean directions inverted through the origin are indicated with a plus symbol.

trend northwest, parallel to turtleback fold axes (Wright and others, 1974). Intrusions ca. 8.7 Ma record 50° to 80° of clockwise rotation (Table 1) but crosscut mylonitic fabrics with the northwest-trending lineation. Rotation of the range core thus appears to have occurred relatively late in its deformational history. If present-day northwest-trending mylonitic lineations reflect early unroofing, then the direction of separation between the Panamint block and the Black Mountains in middle Miocene time would have been approximately east-west.

Deformation Prior To Vertical-Axis Rotation

We restore the data from the younger intrusions into statistical agreement with a reverse-polarity mid-Miocene expected direction, assuming simple, vertical-axis rotation, leaving the data from the 11.6 Ma Willow Spring Pluton to consider. Declinations of site mean directions from the intrusion now cluster around southwest to south-southwest values; the dispersion of inclinations, of course, remains unchanged (Fig. 15). The site mean directions are discordant, in both declination and inclination, from normal and reverse polarity Miocene expected directions, implying deformation of the batholith between crystallization ca. 11.6 Ma and intrusion of the granites ca. 8.7 Ma.

Two possible interpretations for the partially restored data from the Willow Spring Pluton exist, depending on the polarity of magnetization initially acquired by the majority of the sites in the pluton. Both interpretations require differing amounts and sense of tilt, possibly associated with an additional component of deformation that is difficult to fully assess, that is, more local-scale folding or faulting about the currently northwest-trending turtlebacks (Fig. 15). The simplest interpretation assumes an initial reverse polarity for most sites and involves southwest-side–down tilting of some 20° to 40° (Fig. 16a) for the sites with moderate negative polarity. If the near-horizontal mean direction of the entire collection of sites is used for this interpretation, then the amount of tilting increases to about 70°. We interpret the sites with moderate positive inclinations, however, to have undergone additional, local tilting associated with later folding (Fig. 16b), as discussed below. In restored coordinates, the tilt axis would be at a high angle (50° to 80°) to the extension direction suggested by the turtlebacks. Using a normal polarity magnetization, an alternate interpretation involves an improbably large tilt (about 100°–140° for

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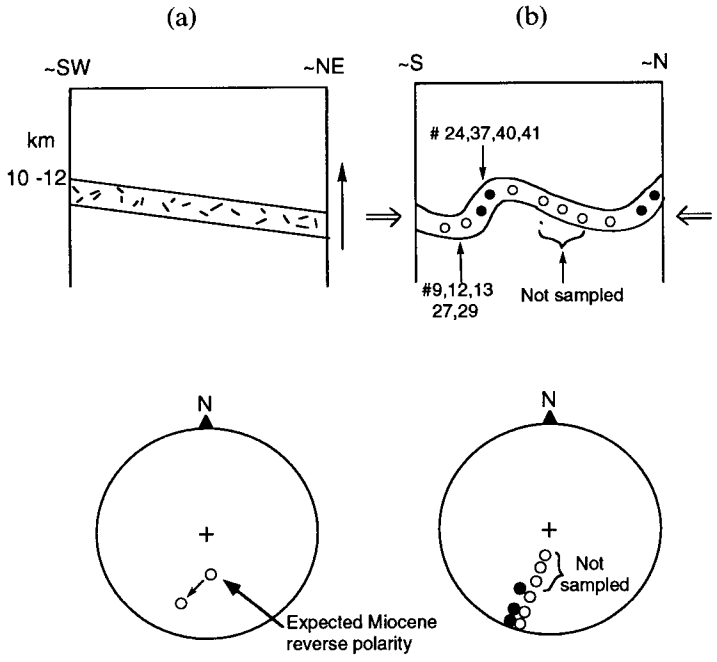


Figure 16. Diagrammatic summary of interpretation for paleomagnetic directions prior to intrusion of younger granitic intrusions and dikes ca. 8.7 Ma. (a) Northeast-side–up tilting of the Willow Spring Pluton, represented as a sill-like body, during onset of unroofing and detachment faulting. (b) Folding of the pluton and initial formation of the Black Mountains turtlebacks resulting in dispersion of site mean directions, particularly inclination values. In schematic equal-area projections, solid (open) symbols represent lower- (upper-) hemisphere projections.

the average of the entire collection) in the mid-crust in a sense of north-northeast side down (Fig. 15) over a short time span (about 2.5 to 3.0 m.y.). We consider the former interpretation to be much more likely, especially considering the size and continuity of field relations of the sampling area. Range-scale southwest-side–down tilting of about 20° to 40° is more consistent with the rarity of tilts greatly exceeding 90° in extensional terranes and with the low average initial dip suggested for the west-dipping detachment fault in the Black Mountains (Holm, 1991; Holm and others, 1992).

Our preferred interpretation of a first phase of deformation of the Black Mountains is complicated by the large variation in inclination data from the Willow Spring Pluton. As briefly noted above, inclinations show a general systematic variation with site locality around the Copper Canyon turtleback fold. We interpret the variation to reflect folding, possibly concurrent with tilting, of both the batholith and the Precambrian rocks making up the Death Valley turtlebacks. Hill and Troxel (1966) were the first to suggest that the Death Valley turtlebacks provide evidence for Tertiary folding. The overall map pattern of the Willow Spring Pluton on a gross scale appears to mimic the folds of the turtlebacks (Fig. 2; see also Pavlis, 1991). In addition, a Miocene hanging-wall sedimentary basin is folded into a northwest-trending syncline with an axial surface coplanar with the axial surface of a synform in the underlying Copper Canyon turtleback fault (Holm and Lux, 1991).

An interpretive cross section of the Willow Spring Pluton (Fig. 16) shows how deformation could result in the observed distribution of paleomagnetic directions prior to vertical-axis rotation (Fig. 15). Soon after emplacement, a gently northeast-dipping pluton (Fig. 16a) tilts 20° to 40°, southwest-side down, presumably associated with early tectonic unroofing. The pluton is subsequently or perhaps simultaneously folded into approximately east-west–trending open antiforms and synforms, increasing dispersion of site mean inclinations (Fig. 16b). Data from the 11 accepted sites do not record the complete dispersion expected, and we interpret this to reflect incomplete sampling around folds (sites shown in Fig. 16b).

Finally, we note that assessment of the dispersion of site mean directions in the Willow Spring Pluton must also recognize the fact that, at least on a local site level, magnetizations of high, narrow laboratory unblocking temperature in the intrusion probably do not adequately average secular variation over several thousands to tens of thousands of years. The magnetization record of sites in intrusions where magnetization blocking occurred over a narrow range of temperature, depending on the cooling history of the intrusion, may be similar to that of individual, rapidly cooled lava flows (Dodson and others, 1978; Haeussler and others, 1992). Thus a contributor to between-site dispersion includes some amount of paleosecular variation and field transitional behavior, as different sites cooled through a narrow range of blocking temperatures over short but different time intervals.

SUMMARY AND IMPLICATIONS

Paleomagnetic data obtained in this study are interpreted to indicate a complex history of deformation of the Black Mountains associated with extensional unroofing. Two generations of Miocene plutons and dikes, separated in time by as little as 3 to 5 m.y., give paleomagnetic directions discordant to one another as well as to expected late Cenozoic reference values. A preferred interpretation of the data, considered in light of thermal and structural information on the unroofing history of the Black Mountains, implies southwest-side–down tilting associated with early unroofing and the onset of folding of the 11.6 Ma Willow Spring Pluton (Figs. 17A, 17B) between 11.6 and 8.7 Ma, followed by the main phase of unroofing (Fig. 17C) between 8.7 and 6.5 Ma, and, lastly, large-scale (50° to 80°) clockwise rotation (Fig. 17D).

The east-west orientation of early-formed folds (11.5 to 8.7 Ma, prior to clockwise rotation) suggests approximately north-south constriction. Folding has continued since 8.7 Ma as evidenced by folded, upper Miocene–lower Pliocene hanging-wall sedimentary strata. The current orientation of the early and late folds is coaxial (northwest-southeast–oriented axial surfaces) in spite of large amounts of rotation. A prolonged history of coaxial shortening is surprising. Rotation of the strain field over time, such that the orientation of constriction exactly "tracks" the rotation of the early fold structures seems highly improbable. Rather, the coaxial deformation might be caused by the early fold structures imparting an anisotropy on the crust. After a fold has nucleated, continued shortening in that orientation might be fa-

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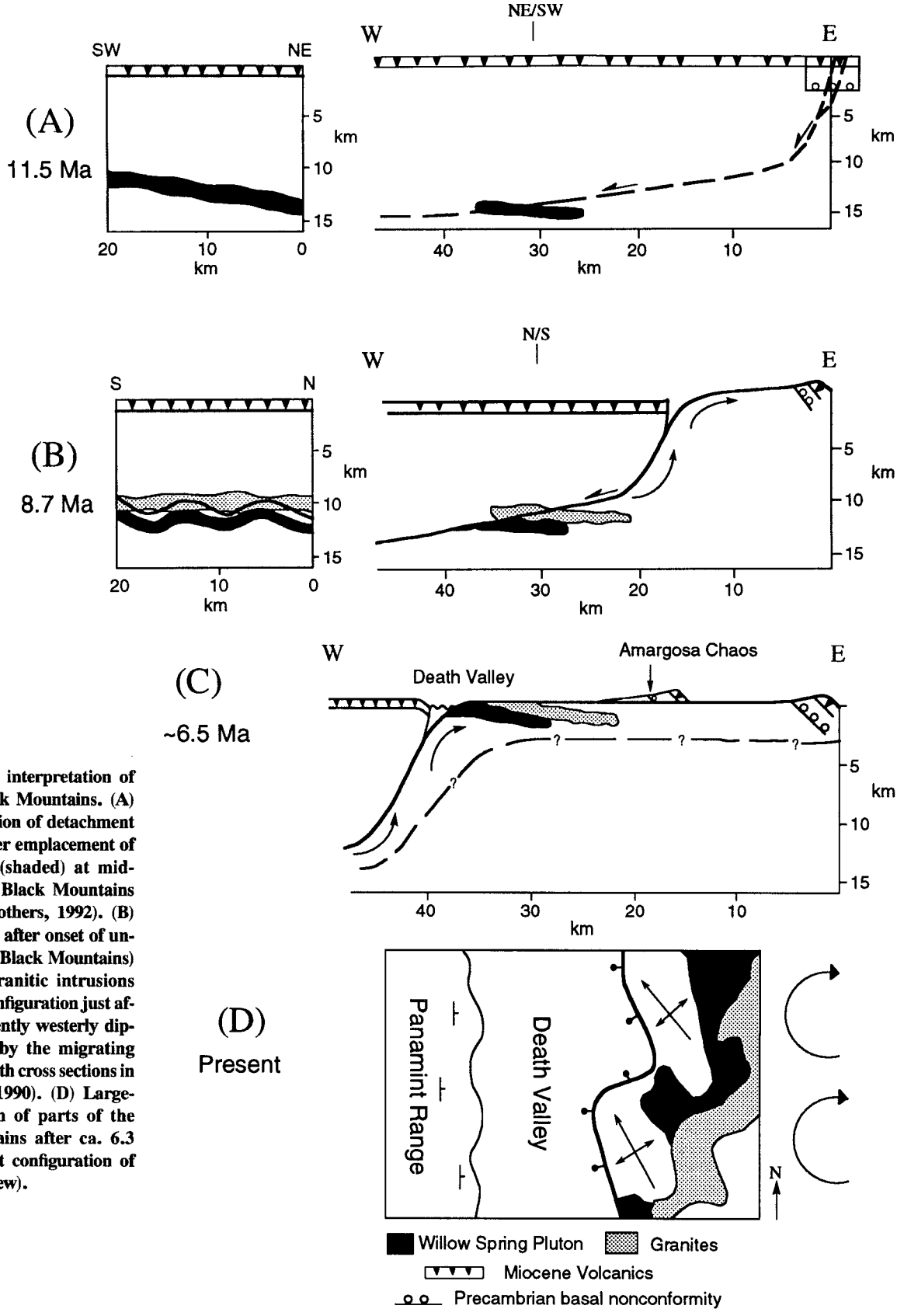


Figure 17. Schematic interpretation of deformation in the Black Mountains. (A) Pre-extension configuration of detachment fault at 11.5 Ma just after emplacement of Willow Spring Pluton (shaded) at mid-crust levels into future Black Mountains block (after Holm and others, 1992). (B) Configuration at 8.7 Ma after onset of unroofing (in southeastern Black Mountains) and emplacement of granitic intrusions (stippled region). (C) Configuration just after unroofing along a gently westerly dipping detachment fault by the migrating hinge model (compare with cross sections in Asmerom and others, 1990). (D) Large-scale clockwise rotation of parts of the northern Black Mountains after ca. 6.3 Ma, resulting in present configuration of the range block (map view).

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vored over the formation of new folds in differing orientations.

We interpret the mechanism of unroofing of the range to be similar to that described in a rolling-hinge model (Buck, 1988; Hamilton, 1988; Wernicke and Axen, 1988). In this model, footwall rocks isostatically rebound as they move out from beneath a scoop-shaped, hanging-wall block (Fig. 17B). The apparent lack of, or only slight inclination discordance in, the paleomagnetic data for the younger intrusions (Fig. 3b), which were emplaced at moderate crustal depths (Holm and others, 1992), is interpreted to indicate little or no net tilting of footwall rocks since 8.7 Ma. This result is consistent with the low angle of initial dip of the detachment proposed by Holm and others (1992). At the depths and time at which the plutons were intruded (8 to 15 km at 11.6 to 8.7 Ma), Holm (1991) argued for a detachment dip of only 6° to 13°. This would allow a maximum net tilt associated with unroofing of only about 10°, considering that the detachment is currently subhorizontal. An apparent lack of tilt does not preclude tilting greater than 10° during unroofing. In the rolling-hinge model, footwall rocks which tilt at the onset of unroofing will untilt as the flexure migrates through the upper crust. In spite of possibly large amounts of tilt during unroofing, the net result would be little or no tilt shown by the paleomagnetic data. The migration of the footwall unroofing, however, would still be recorded by a progression of decreasing cooling ages in the direction of hanging-wall transport, or northwest in present-day coordinates (Holm and others, 1992).

The paleomagnetic results are further interpreted to suggest that large-scale clockwise rotation of the crystalline core occurred late in the deformational history. Because this study focused principally on young intrusions in the northwest portion of the range, we do not know whether the entire range rotated or if rotation was restricted to the area sampled in this study. Rotation of the entire range (as proposed for other range blocks north of the Black Mountains) would seem to require the presence of a separate, structurally lower detachment fault beneath the Black Mountains. Although we cannot rule out this possibility, we believe that this is unlikely considering that no large-scale detachment fault of post–8.7 Ma age has been identified east of the Black Mountains.

The clockwise bend in the range of about 45° at its northwestern end (Fig. 1) suggests that rotation might indeed be a more localized phenomenon. This would suggest that rotation is related to right-lateral shearing of the Death Valley fault zone analogous to that seen for the Las Vegas Range adjacent to the Las Vegas Valley shear zone (Gillett and Van Alstine, 1982). The presence of miogeoclinal strata in the Las Vegas Range striking at a high angle to the fault zone facilitates recognition of oroflexure adjacent to the Las Vegas Valley shear zone. The lack of stratified rocks in the core of the Black Mountains hinders identification of such a feature.

Large-scale rotation of diverse magnitude and sense associated with Tertiary extension may have affected other parts of the Death Valley region (for example, Grapevine and Bare Mountain; Snow and Wernicke, 1989) and may be a fundamental characteristic of strongly extended continental lithosphere. Although ductile stretching lineations within mylonites of metamorphic core complexes have been used to assess paleoextension directions (Wust, 1986), the interpretations assume little or no vertical-axis rotation since formation of these structures. Given the evidence for late-stage heterogeneous rotation in the Death Valley area and in other highly extended regions, it seems probable that such an assessment is an oversimplication and that more rigorous evaluation, where possible, is necessary.

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